

Analyzing GPS Satellite Visibility and Dilution of Precision (DOP) to Evaluate Positioning Errors

1. Abstract

This project investigates how GPS satellite visibility and satellite geometry impact Dilution of Precision (DOP) and, consequently, the accuracy of GPS positioning. Through an analysis of satellite configurations and visibility, this research quantifies the effect of DOP on GPS errors, providing insights into conditions for optimal GPS accuracy. By gathering data on satellite visibility and calculating DOP values, this project aims to identify specific configurations that minimize GPS errors, offering practical applications for improving GPS reliability across varied environments.

2. Introduction

2.1. Background

GNSS technology [3] has become an integral part of modern life, providing essential positioning and navigation services in a wide range of applications. From transportation and telecommunications to precision mapping and even autonomous vehicles, GPS is critical to our technological infrastructure.

GPS operates by trilaterating signals [1] from a constellation of satellites orbiting the Earth. Each satellite transmits its location and precise time data to GPS receivers, which use this information to calculate their own position, and time. However, the accuracy of this calculation depends on several factors, particularly the number of satellites in view and their spatial arrangement relative to the receiver. These factors contribute to what known as Dilution of Precision (DOP), a measure of satellite geometry quality and its influence on GPS accuracy. Dilution of precision (DOP), or geometric dilution of precision (GDOP), is a term used in satellite navigation and geomatics engineering to specify the error propagation as a mathematical effect of navigation satellite geometry on positional measurement precision.

2.2.Problem Statement

Despite its accuracy, GPS is not error-proof. Variability in satellite visibility and geometry can significantly degrade GPS accuracy, especially in environments where satellite signals are blocked by buildings, mountains, or trees, Poor satellite configuration can lead to high DOP values, amplifying GPS errors [7]. Understanding how visibility and DOP impact GPS positioning accuracy is critical for applications where precision is essential. This project addresses these challenges by examining the relationship between satellite visibility, DOP, and positioning errors to provide insights that could improve GPS accuracy in diverse conditions.

3. Theoretical Insights and Practical Applications [2]

GPS systems depend on signals transmitted by a constellation of satellites in orbit, which interact with terrestrial receivers to determine precise positioning. The performance of these systems is inherently influenced by the visibility of satellites and the spatial geometry they form relative to the receiver.

Theoretical Foundations

1. Trilaterating and Satellite Geometry:

GPS receivers rely on signals from at least four satellites to calculate a three-dimensional position (latitude, longitude, and altitude) and time offset. The relative spatial arrangement of satellites, known as satellite geometry, critically determines the accuracy of this triangulation process. The quality of satellite geometry is quantified using metrics such as DOP. Low DOP values indicate a favorable geometry that minimizes positioning errors, while high DOP values, arising from poor satellite configuration, exacerbate errors.

- **Factors Affecting Satellite Visibility:** Satellite visibility is influenced by several dynamic and static factors; Orbital Mechanics, GPS satellites follow predictable orbits, but their visibility depends on the observer's geographic location, time, and environmental conditions. Atmospheric Conditions, Ionospheric and tropospheric disturbances can degrade signal integrity, adding to the positional error.

2. Dilution of Precision (DOP) Metrics:

DOP is subdivided into several components, each addressing specific aspects of positional accuracy. Geometric DOP (GDOP), a comprehensive measure encompassing spatial and temporal errors. Horizontal and Vertical DOP (HDOP and VDOP), indicators of error

magnitude in the horizontal plane and vertical axis, respectively. Position DOP (PDOP), Reflects three-dimensional positional accuracy [8]. Time DOP (TDOP), Relates to the precision of the timing solution.

The equation for GDOP is derived from the Geometry Matrix G , which relates satellite positions to the receiver's position. Given G , the GDOP is calculated as follows:

$$GDOP = \sqrt{\text{trace}((G^T G)^{-1})} [6]$$

Explanation:

1. Geometry Matrix G :

- G is a $N \times 4$ matrix, where N is the number of visible satellites.
- x_i, y_i, z_i Components of the unit vector (line-of-sight) pointing from the receiver to the i -th satellite.
- The last column is a constant (e.g., 1) that incorporates clock bias.

$$G = \begin{bmatrix} x_i & y_i & z_i & 1 \\ x_{i+1} & y_{i+1} & z_{i+1} & 1 \\ \vdots & \vdots & \ddots & \vdots \\ x_N & y_N & z_N & 1 \end{bmatrix}$$

2. Covariance Matrix:

- The covariance matrix of the receiver's position estimation error is proportional to $(G^T G)^{-1}$

3. Trace:

- The trace of $(G^T G)^{-1}$ is the sum of the diagonal elements, representing the cumulative effect of satellite geometry on position accuracy.

4. GDOP:

- Taking the square root of the trace gives the GDOP value.

Analyzing and Mitigating DOP Impacts

This project investigates how satellite visibility and geometry influence DOP values and, consequently, positioning errors. Key aspects include:

- **Empirical Data Collection:** Satellite visibility is tracked across varied environments to assess how obstruction and satellite configurations affect DOP
- **Quantitative Analysis:** Statistical and computational models are employed to correlate DOP values with observed positional errors.

- **Optimization Strategies:** Recommendations are proposed to enhance satellite configurations and receiver algorithms for improved GPS accuracy in challenging settings.

4. GPS Satellite Visibility and Data Acquisition

In the field of satellite communication and navigation, the ability to predict and analyze satellite visibility is a key factor in ensuring optimal system performance. One of the primary inputs for such analyses is almanac data which provides essential information about the position, trajectory, and operational status of GPS satellites. This chapter outlines the acquisition of Standard Ephemeris Message (SEM) almanac files, their role in GPS satellite visibility analysis, and how these files are used within MATLAB for calculations, visualization, and determining DOP errors.

4.1 Understanding SEM Almanac Files

SEM (Standard Ephemeris Message) files are structured data sets used to represent the position and trajectory of GPS satellites over a certain period [5]. These files are a compact format, enabling efficient processing and storing of orbital parameters such as; Orbital Elements, these define the satellite's orbit, including inclination, eccentricity, and semi-major axis. Clock Corrections, the adjustments necessary to account for the satellite's clock discrepancies.

```

31 CURRENT.ALM
298 503808

1
80
0
2.57492065429688E-04 5.47981262207031E-03 -2.61570676229894E-09
5.15366601562500E+03 -5.28235316276550E-01 2.70691514015198E-01
-4.18479800224304E-01 0.00000000000000E+00 0.00000000000000E+00
63
12

```

Figure 1 SEM Almanac File First Label

Almanac Information:

File: CURRENT.ALM

Week Number: 298

Time of Week (TOW): 503808

Satellite Data:

Satellite Number: 1

Health Status: 80

Reserved Field: 0

Eccentricity: 2.57492065429688E-04

Inclination Angle: 5.47981262207031E-03

Rate of Change of Inclination: -2.61570676229894E-09

Semi-Major Axis: 5153.66601562500

Rate of Change of Right Ascension: -5.28235316276550E-01

Argument of Perigee: 2.70691514015198E-01

Mean Anomaly: -4.18479800224304E-01

Clock Drift Rate: 0.00000000000000E+00

Clock Drift Offset: 0.00000000000000E+00

Reserved Field 2: 63

Reserved Field 3: 12

4.2 Sources for SEM Almanac Files

Obtaining accurate and up-to-date SEM almanac files is crucial for precise satellite tracking. Several trusted repositories and institutions provide these files, and they are freely accessible for research and development. Notable sources include:

- **CelesTrak:** CelesTrak is one of the most widely used sources for satellite data. It provides SEM almanac files in an easy-to-download format. The data is updated regularly, ensuring that users have access to the most current satellite information. [5]

- **US Coast Guard Navigation Center (NAVCEN):** NAVCEN offers a comprehensive repository of GPS almanac data. These files are updated and can be used for precise positioning and satellite visibility calculations. [4]
- **International GNSS Service (IGS):** The IGS is an international body provides GNSS- related data, including ephemeris and almanac files. These are often used for high-accuracy scientific research and applications. [3]

4.3 Using SEM Files in MATLAB for satellite Visibility

Once an SEM almanac file has been acquired, it can be used in MATLAB for satellite visibility analysis [6]. MATLAB's Satellite Communications Toolbox provides functions for integrating this data into satellite scenarios. The workflow for utilizing SEM files is outlined below:

- **Define Parameters and Time Setup:**

Set the observation time parameters, including the start time, stop time, and sampling interval. This defines the time range for visibility analysis.

```
% Define Observation Time
queryTime = datetime('now', TimeZone="UTC");
startTime = queryTime - hours(12);
stopTime = queryTime + hours(13);
sampleTime = 60; % seconds
```

- **Create a Satellite Scenario:**

Use the “satelliteScenario” function to set up the parameter’s visibility, including the time frame and satellite configuration.

```
% Create Satellite Scenario
sc = satelliteScenario(startTime, stopTime, sampleTime);
```

- **Define Ground Stations for Different Environments:**

Define the ground station’s location using its latitude, longitude, and minimum elevation angle for satellite visibility. This ensures that only satellites above a certain elevation are considered visible.

```
% Define ground stations for different environments
areas = [
    -3.4653, -62.2159, 0; % Amazon Rainforest (Forest Area)
    40.7128, -74.0060, 0; % Manhattan, New York (Urban Area)
    23.4162, 25.6628, 0; % Sahara Desert (Non-Urban Area)
];
areaLabels = {'Forest Area', 'Urban Area', 'Non-Urban Area'};
elevationMasks = [30, 25, 10]; % Degrees of minimum elevation angle for each area
```

- **Import the SEM Almanac File:**

The “satellite” function is then used to load the SEM almanac file, providing the necessary satellite orbital data for the scenario.

```
% Add GPS Satellite Constellation (use SEM Almanac)
semAlmanacFileName = "D:\Ders\Master\Semester1\Attitude and Navigation Systems\SEM
almanac\almanac.sem.week0296.405504.txt";
sat = satellite(sc, semAlmanacFileName)
```

- **Access Analysis and Visibility Determination:**

For each ground station, calculate satellite visibility based on line-of-sight constraints and minimum elevation angles. Store azimuth and elevation data for visible satellites.

```
% Initialize variables to store GDOP values and position errors
gdopValues = [];
posErrors = []; % Hypothetical position errors for plotting

for i = 1:length(areas)
    % Extract latitude, longitude, altitude
    lat = areas(i, 1);
    lon = areas(i, 2);
    alt = areas(i, 3);
    minElevation = elevationMasks(i); % Set elevation mask based on environment

    % Perform access analysis
    azimuths = [];
    elevations = [];
    gdopForArea = [];

    % Collect satellite visibility data and calculate GDOP
    for j = 1:length(sat)
        [az, el, dist] = aer(gs, sat(j), queryTime);
        if el > minElevation % Only include satellites above the mask
            azimuths(end+1) = az;
            elevations(end+1) = el;
        end
    end
end
```

- **Calculate DOP Based on Satellite Geometry**

Use the geometric distribution of visible satellites to calculate GDOP for each ground station. This step involves creating a geometry matrix and checking its condition number.

```
% Calculate GDOP if sufficient satellites are visible
if length(azimuths) >= 4
    % Generate Geometry Matrix for all visible satellites
    G = zeros(length(azimuths), 4);
    for k = 1:length(azimuths)
        G(k, :) = [
            cosd(elevations(k)) * cosd(azimuths(k)), ...
            cosd(elevations(k)) * sind(azimuths(k)), ...
```

```

        sind(elevations(k)), 1
    ];
end

% Check matrix condition and compute GDOP
if cond(G' * G) < 1e10
    gdop = sqrt(trace(pinv(G' * G)));
    gdopForArea = [gdopForArea, gdop];
else
    fprintf('Matrix for %s is ill-conditioned. GDOP cannot be computed.\n',
areaLabels{i});
end
else
    fprintf('Insufficient satellites visible in %s.\n', areaLabels{i});
end
end

```

- **Visualization of Results:**

Once the visibility is complete, the results can be visualized in MATLAB. The “skyplot” function can be used to generate a sky plot showing the azimuth and elevation of the visible satellites. Additionally, satellite positions can be animated over time to observe their movement and visibility dynamically.

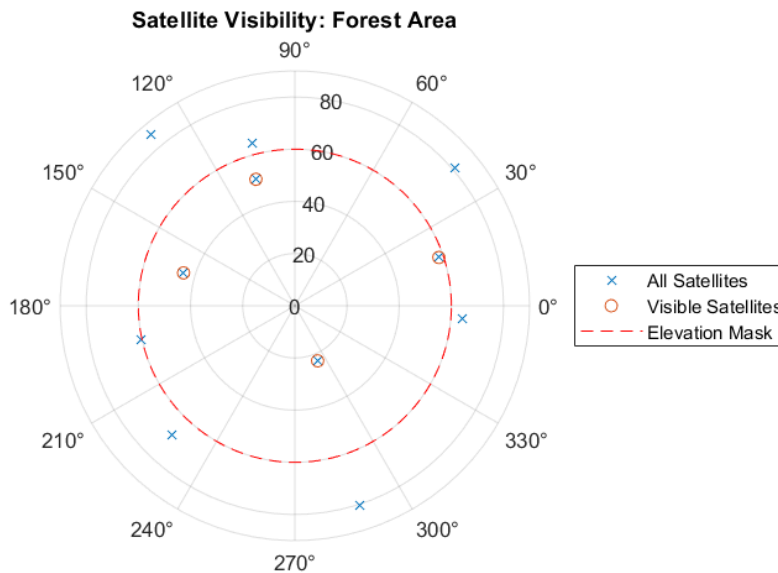


Figure 2 Satellite Visibility for Forest Area

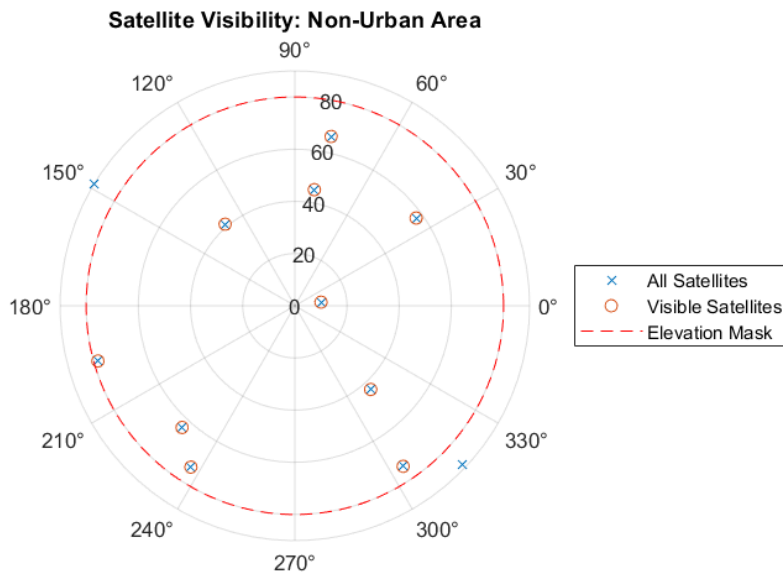


Figure 3 Satellite Visibility for Non-Urban Area

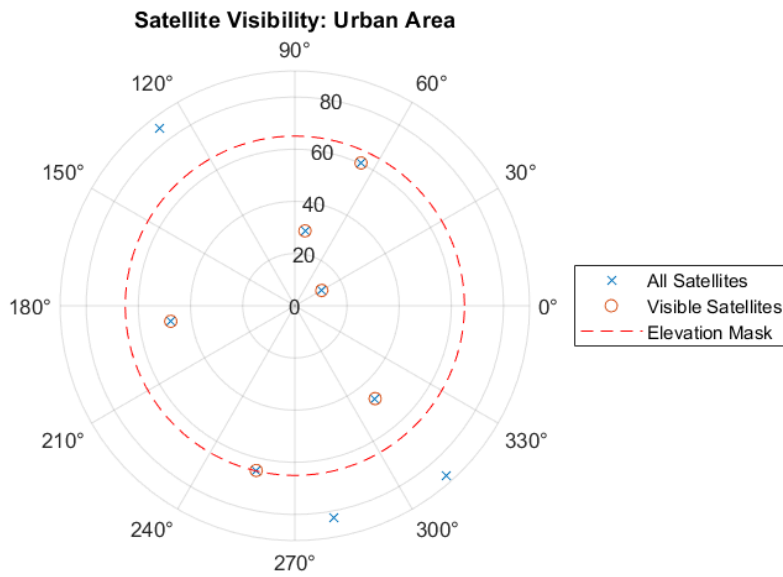


Figure 4 Satellite Visibility for Urban Areas

- **Matrix G Real Values**

	1	2	3	4
1	-0.6087	-0.6570	0.4448	1
2	-0.5198	-0.8062	0.2828	1
3	0.1740	0.0241	0.9844	1
4	0.6823	0.4935	0.5394	1
5	0.4605	-0.5080	0.7279	1
6	0.1172	0.6985	0.7060	1
7	-0.4274	0.4997	0.7534	1
8	-0.9433	-0.2645	0.2005	1
9	0.5382	-0.7978	0.2718	1
10	0.1919	0.8955	0.4015	1

Figure 5 G Matrix for GDOP Calculation

- **GDOP Values After the Calculation**

GDOP for Forest Area: 10.08

GDOP for Urban Area: 3.01

GDOP for Non-Urban Area: 1.91

5. DOP Error Analysis

To explore the relationship between satellite geometry and positional accuracy, this study focuses on calculating Geometric Dilution of Precision (GDOP) values across different environmental scenarios and visualizing satellite visibility. The GDOP metric is derived based on the geometric distribution of visible satellites, which directly impacts the accuracy of position estimation. Using predefined elevation masks and satellite data, the analysis evaluates visibility conditions in three distinct environments: forest areas, urban settings, and non urban regions.

For each environment, satellite visibility charts were generated, with the observer at the center point looking outward, to illustrate the spatial distribution of satellites above the horizon. Additionally, GDOP values were calculated for scenarios with varying satellite availability. These values are plotted against hypothetical positional errors to demonstrate the general impact of GDOP on position accuracy.

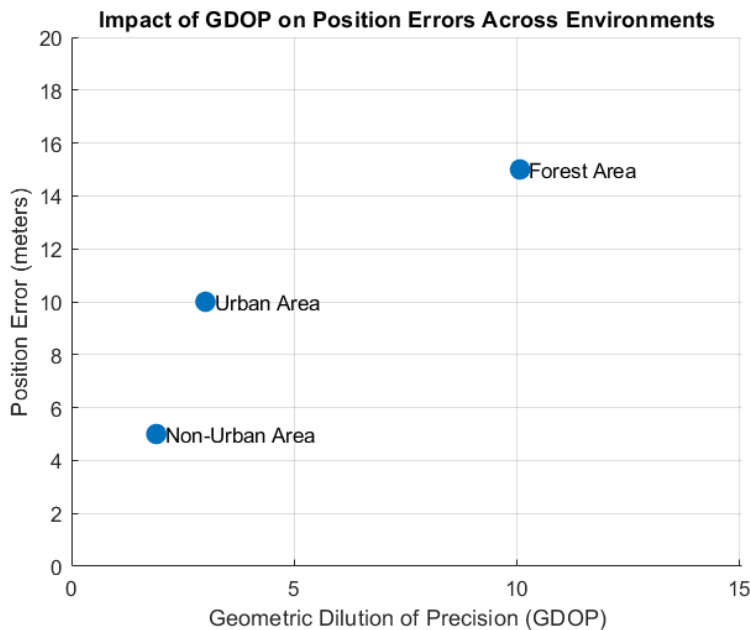


Figure 6 GDOP Values-Position Error Graph

Conclusion

To explore the relationship between satellite geometry and positional accuracy, this study focuses on calculating Geometric Dilution of Precision (GDOP) values across different environmental scenarios and visualizing satellite visibility. The GDOP metric is derived based on the geometric distribution of visible satellites, which directly impacts the accuracy of position estimation. Using predefined elevation masks and satellite data, the analysis evaluates visibility conditions in three distinct environments: forest areas, urban settings, and open desert regions.

For each environment, satellite visibility charts were generated, with the observer at the center point looking outward, to illustrate the spatial distribution of satellites above the horizon. Additionally, GDOP values were calculated for scenarios with varying satellite availability. These values are plotted against hypothetical positional errors to demonstrate the general impact of GDOP on position accuracy.

Additionally, the integration of SEM almanac files into MATLAB facilitated precise calculations and visualizations, providing a robust framework for real-world applications. The correlation between DOP metrics and positional errors underscores the need for strategic satellite placement and advanced receiver algorithms to mitigate positioning inaccuracies in challenging environments.

In summary, this project emphasizes the importance of understanding and analyzing Geometric Dilution of Precision (GDOP) to evaluate the reliability and accuracy of GPS systems. By calculating GDOP values for different environments and satellite configurations, the study demonstrates how satellite visibility and geometric alignment significantly impact positioning performance.

APPENDIX

Satellite Visibility and Error Analysis Source Codes

```
%% 1. Define Parameters and Time Setup
queryTime = datetime('now', TimeZone="UTC");
startTime = queryTime - hours(12);
stopTime = queryTime + hours(13);
sampleTime = 60; % seconds

%% 2. Define Ground Stations for Different Environments
areas = [
    -3.4653, -62.2159, 0; % Amazon Rainforest (Forest Area)
    40.7128, -74.0060, 0; % Manhattan, New York (Urban Area)
    23.4162, 25.6628, 0; % Sahara Desert (Non-Urban Area)
];
areaLabels = {'Forest Area', 'Urban Area', 'Non-Urban Area'};
elevationMasks = [30, 25, 10]; % Degrees of minimum elevation angle for each area

%% 3. Load Satellite Data
semAlmanacFileName = "D:\\Ders\\Master\\Semester1\\Attitude and Navigation
Systems\\SEM almanac\\almanac.sem.week0301.589824.txt";

%% 4. Initialize Variables to Store Results
gdopValues = [];
posErrors = [];

%% 5. Loop Through Each Area to Analyze GDOP
for i = 1:length(areas)
    % Extract latitude, longitude, altitude
    lat = areas(i, 1);
    lon = areas(i, 2);
    alt = areas(i, 3);
    minElevation = elevationMasks(i); % Set elevation mask based on environment

    % Create satellite scenario
    sc = satelliteScenario(startTime, stopTime, sampleTime);
    sat = satellite(sc, semAlmanacFileName);

    % Define ground station with specific elevation mask
    gs = groundStation(sc, lat, lon, MinElevationAngle=minElevation);

    % Perform access analysis and calculate GDOP
    allAzimuths = [];
    allElevations = [];
    visibleAzimuths = [];
    visibleElevations = [];

    for j = 1:length(sat)
        [az, el, dist] = aer(gs, sat(j), queryTime);

        % Store all satellite data
        allAzimuths(end+1) = az;
        allElevations(end+1) = el;

        % Filter satellites above the minimum elevation mask
    end
end
```

```

        if el > minElevation
            visibleAzimuths(end+1) = az;
            visibleElevations(end+1) = el;
        end
    end

    % Calculate GDOP if sufficient satellites are visible
    if length(visibleAzimuths) >= 4
        % Generate Geometry Matrix for all visible satellites
        G = zeros(length(visibleAzimuths), 4);
        for k = 1:length(visibleAzimuths)
            G(k, :) = [
                cosd(visibleElevations(k)) * cosd(visibleAzimuths(k)), ...
                cosd(visibleElevations(k)) * sind(visibleAzimuths(k)), ...
                sind(visibleElevations(k)), 1
            ];
        end

        % Calculate GDOP using trace method
        GDOP = sqrt(trace(inv(G' * G)));
        gdopValues(end+1) = GDOP;
        fprintf('GDOP for %s: %.2f\n', areaLabels{i}, GDOP);
    else
        fprintf('Insufficient satellites visible in %s.\n', areaLabels{i});
        gdopValues(end+1) = NaN; % No valid GDOP
    end

    % Assign hypothetical position errors for plotting
    posErrors(end+1) = 5 * (4 - i);

    % Plot Satellite Visibility Chart
    figure;
    polarplot(deg2rad(allAzimuths), 90 - allElevations, 'x', 'DisplayName', 'All
Satellites');
    hold on;
    polarplot(deg2rad(visibleAzimuths), 90 - visibleElevations, 'o', 'DisplayName',
'Visible Satellites');

    % Add mask boundary for visualization
    theta = linspace(0, 2*pi, 360); % Full circle for elevation mask
    maskRadius = 90 - minElevation; % Convert elevation angle to polar radius
    polarplot(theta, repmat(maskRadius, size(theta)), '--r', 'DisplayName',
'Elevation Mask');

    % Customize plot appearance
    title(sprintf('Satellite Visibility: %s', areaLabels{i}));
    legend('show');
    rlim([0, 90]); % Elevation range from 0° to 90°
    hold off;
end

%% 6. Plot GDOP vs Position Error
figure;
scatter(gdopValues, posErrors, 100, 'filled');

% Annotate points with area labels

```

```

for i = 1:length(areaLabels)
    text(gdopValues(i) + 0.2, posErrors(i), areaLabels{i}, 'FontSize', 10);
end

% Add labels, title, and grid
xlabel('Geometric Dilution of Precision (GDOP)');
ylabel('Position Error (meters)');
title('Impact of GDOP on Position Errors Across Environments');
grid on;

% Adjust axes for better visibility
xlim([0, max(gdopValues) + 5]);
ylim([0, max(posErrors) + 5]);

```

Bibliography

- [1] Kaplan, E. D., & Hegarty, C. (2006). Understanding GPS: Principles and Applications. Artech House.
- [2] Misra, P., & Enge, P. (2011). Global Positioning System: Signals, Measurements, and Performance. Ganga-Jamuna Press.
- [3] International GNSS Service (IGS). (2023). IGS Products for GNSS Data. Retrieved from <https://igs.org>
- [4] US Coast Guard Navigation Center (NAVCEN). (2023). GPS Almanac Data. Retrieved from <https://www.navcen.uscg.gov>
- [5] CelesTrak. (2023). Standard Ephemeris Message (SEM) Data. Retrieved from <https://celestrak.com>
- [6] MathWorks. (2023). Satellite Communications Toolbox User Guide. MATLAB Documentation. Retrieved from <https://mathworks.com>
- [7] Parkinson, B. W., & Spilker, J. J. (1996). Global Positioning System: Theory and Applications, Volume I & II. American Institute of Aeronautics and Astronautics.
- [8] Hoffmann-Wellenhof, B., Lichtenegger, H., & Collins, J. (2008). Global Navigation Satellite Systems: GPS, GLONASS, Galileo, and More. Springer.